

Leading-Edge Airfoil Modifications for Low- Speed Aerodynamics

ASE Senior Capstone Project



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Problem Statement

- This project tested traditional vs. modified sinusoidal leading-edges (LE).
- Simulations included traditional and sinusoidal LE's, with and without propellers attached.
- Analysis revealed both improvements and unexpected aerodynamic effects.

Objectives

- Design six different LE configurations
 - Clean wing, wing with motor, wing with motor and propellers
- Evaluate performance under varying Reynolds numbers, RPM, and Angle of Attacks
- Determine the most effective setup at low-speed performance

Background

- **Where did this idea come from?**
 - Inspired by biomimicry (whale fins, owl wings), which are great at delaying stall and improving lift.
- **Why study this?**
 - Enhancing low-speed flight can improve efficiency and reduce noise. Also, there have been very few studies done with propellers attached.
- **What is the goal?**
 - Investigate the differences in traditional and sinusoidal leading-edge designs with and without propellers at various conditions.

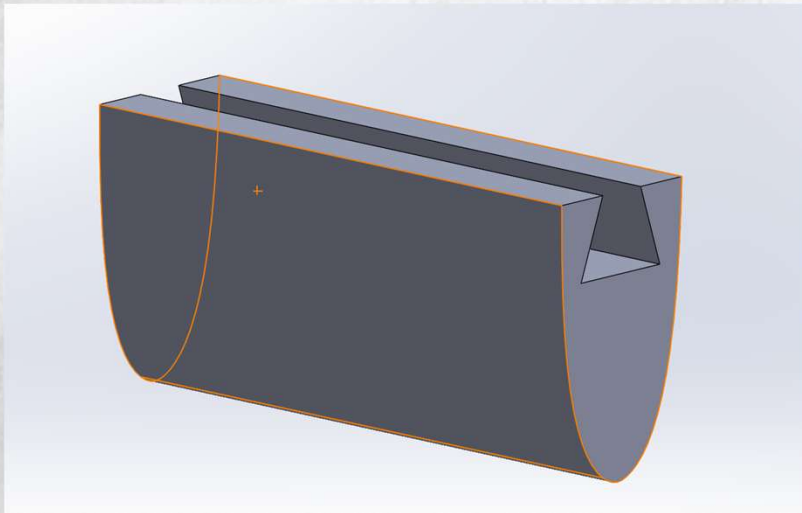


Flipped Flight (J.Pickrell, 2004)

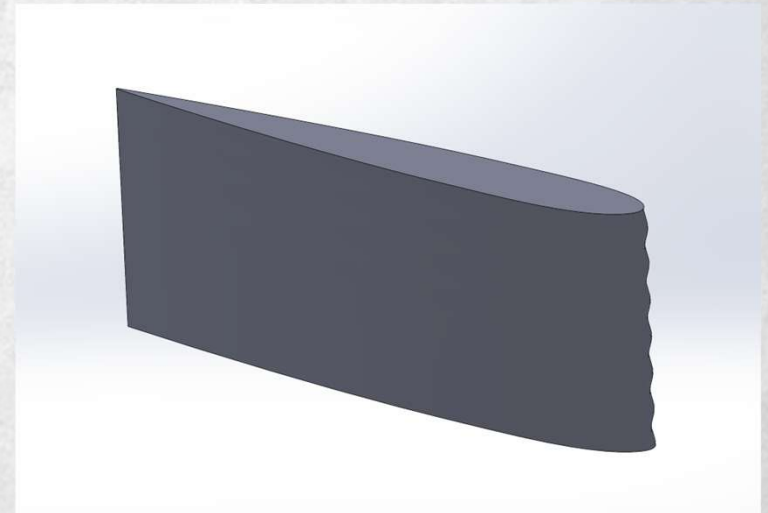


Research/Design Process

- Created NACA0012 airfoils in SolidWorks.
- Modified designs and 3D printed PLA plastic prototypes.



Traditional LE Piece



Sinusoidal LE on Airfoil

Manufacturing/Testing Process



- Used a sliding attachment mechanism for quicker assembly/disassembly.
- Wind tunnel tests: -25° to 40° AOA, 100k–400k Reynolds number, and RPM of 8000–14000.



Load Plate and Rod Manufacturing



Electronic Lathe



Milling Machine



Airfoil with Rods



Traditional Lathe

- Drilled holes into load plate.
 - Holes created for steel rods, load cell, and area for wiring
- Modified steel rods to improve support.
- Adjust airfoil and rod to fit in tunnel dimensions.



Wiring



- Terminated the leads from the power wires.
- Wires were routed through center of the airfoil.
- Connected to corresponding terminals to power each motor.



Wind Tunnel Setup

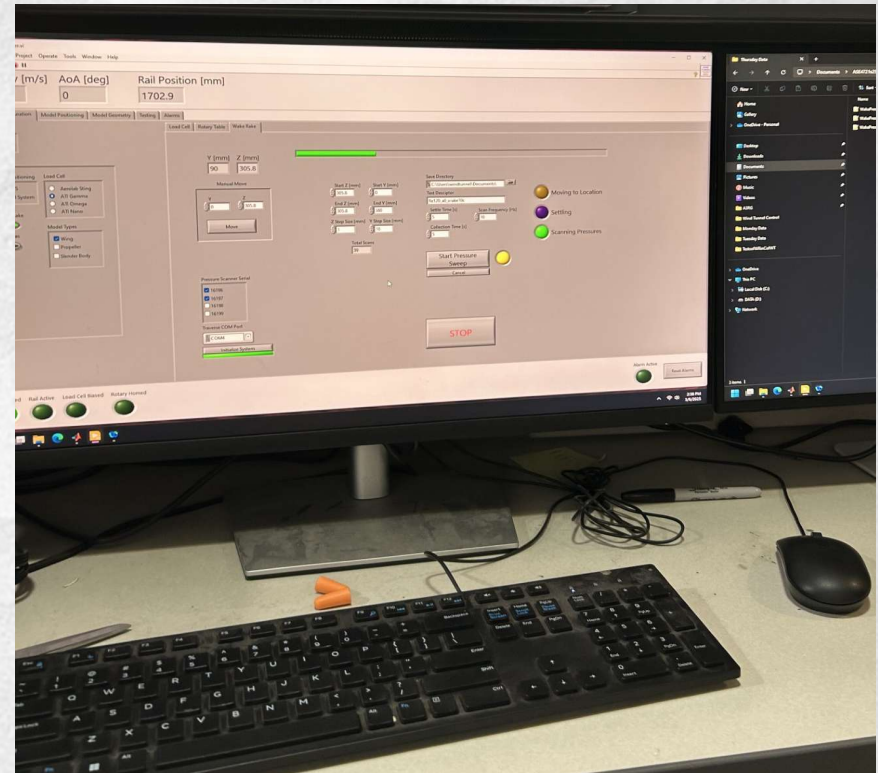


Patterson Wind Tunnel

Dimensions of Test Section:

28" x 40" x 160"

Closed Circuit



LabVIEW Setup



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Clean Wings



Traditional Smooth Edge



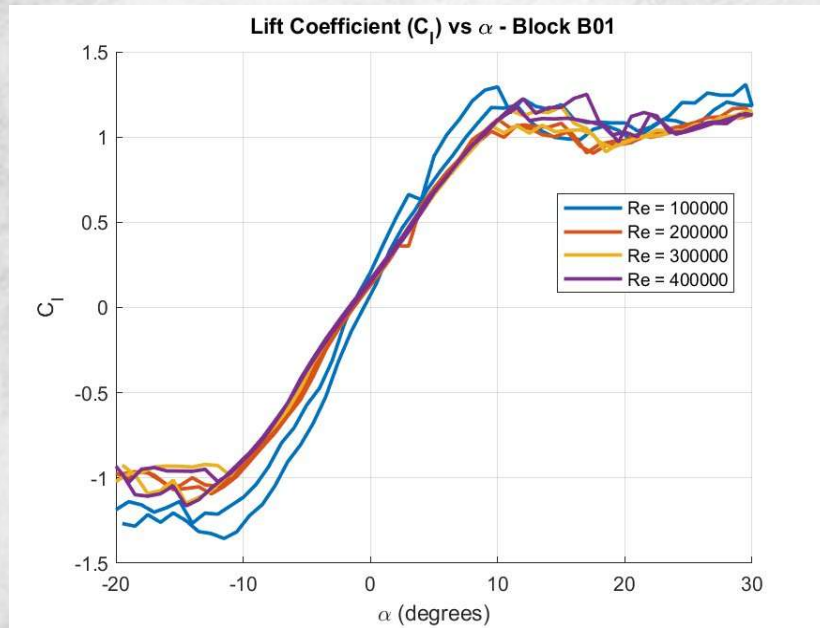
Sinusoidal Edge



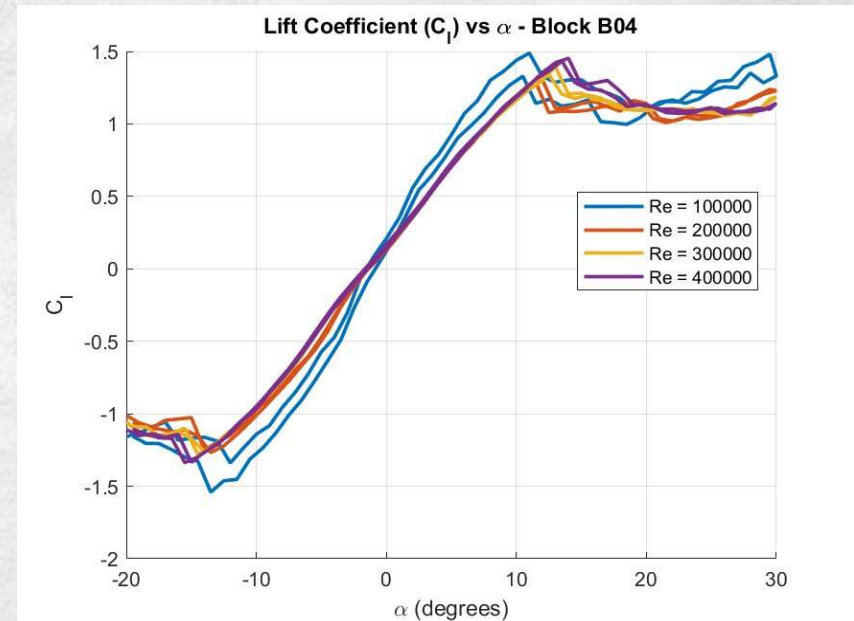
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Results for Cl Clean Wing

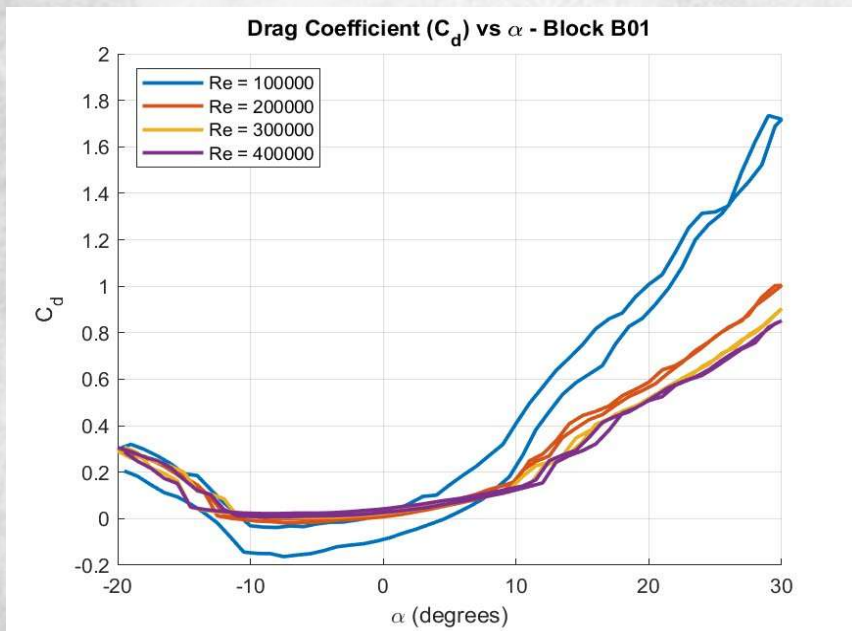


Traditional Leading Edge

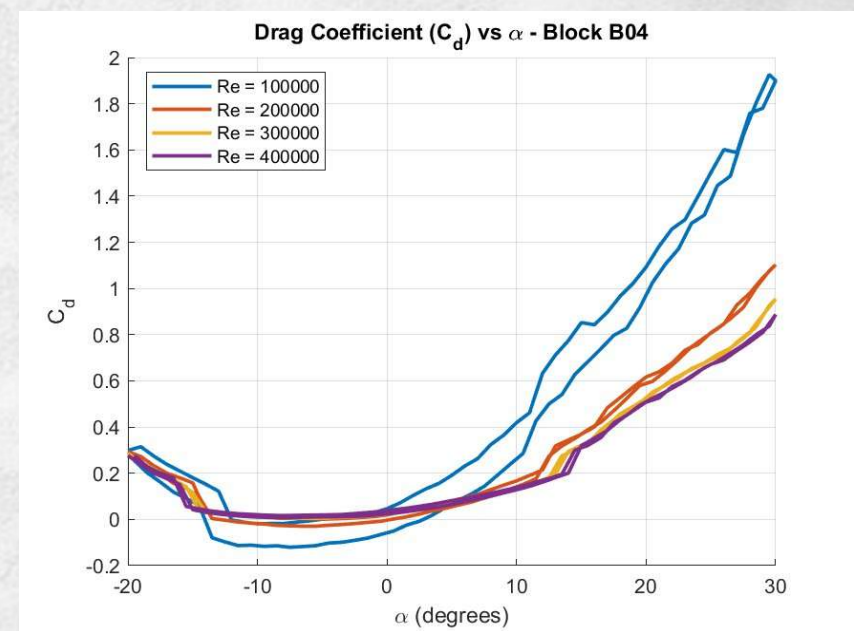


Sinusoidal Leading Edge

Results for C_d Clean Wing

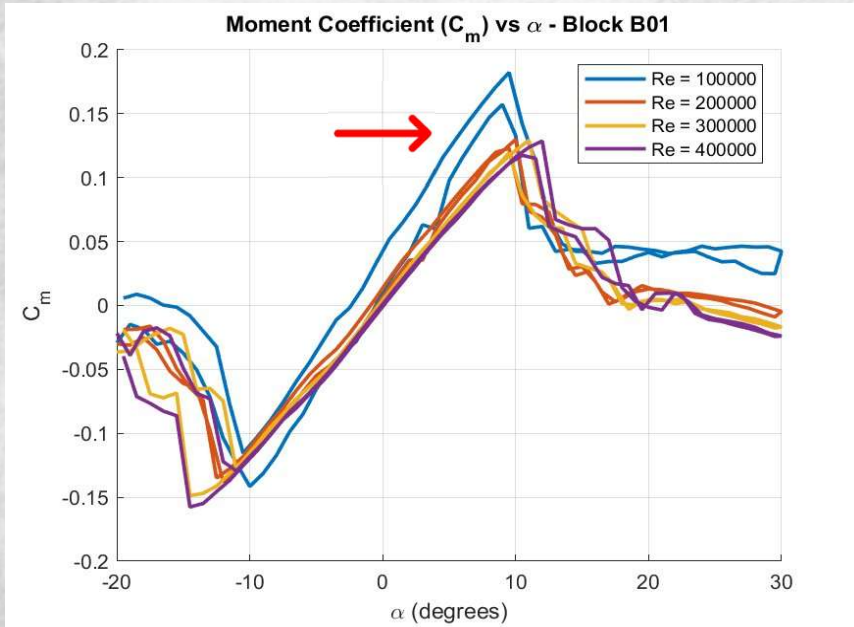


Traditional Leading Edge

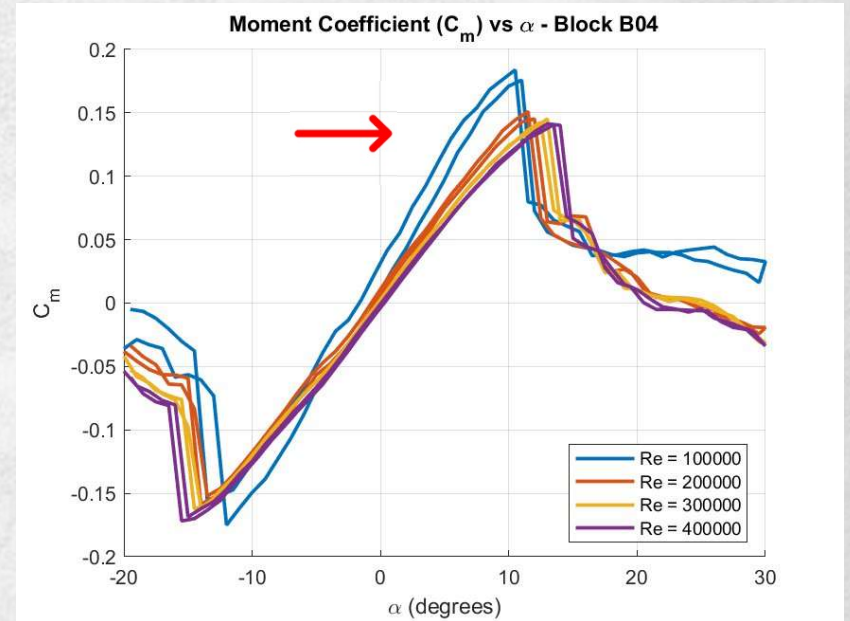


Sinusoidal Leading Edge

Results for C_m Clean Wing



Traditional Leading Edge

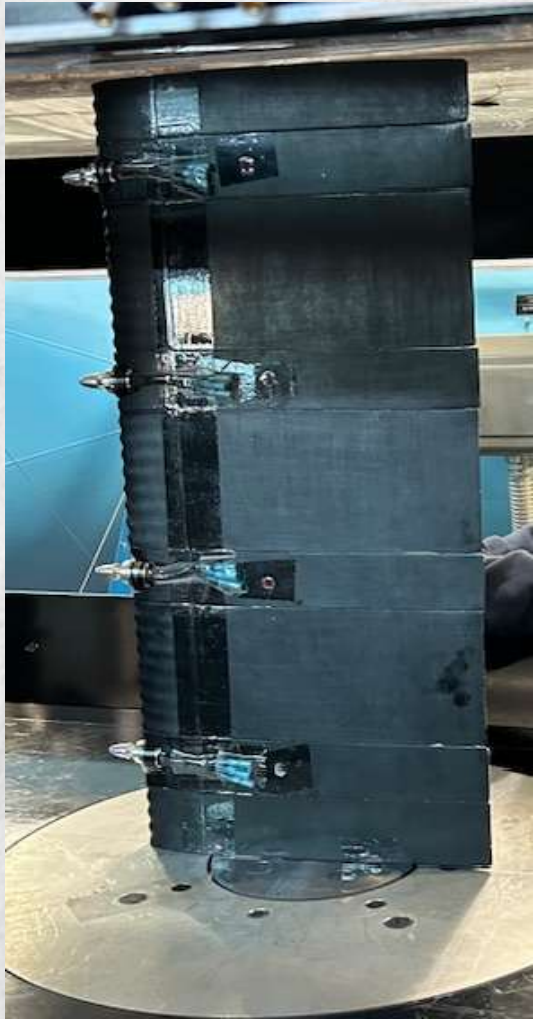


Sinusoidal Leading Edge

Observations of Clean Wing

- Sinusoidal LE showed higher lift values and better efficiency in comparison to the clean leading edge.
- Setup was stable with minimal vibration throughout wind tunnel testing.
- Preliminary data supports leading-edge modification benefits.

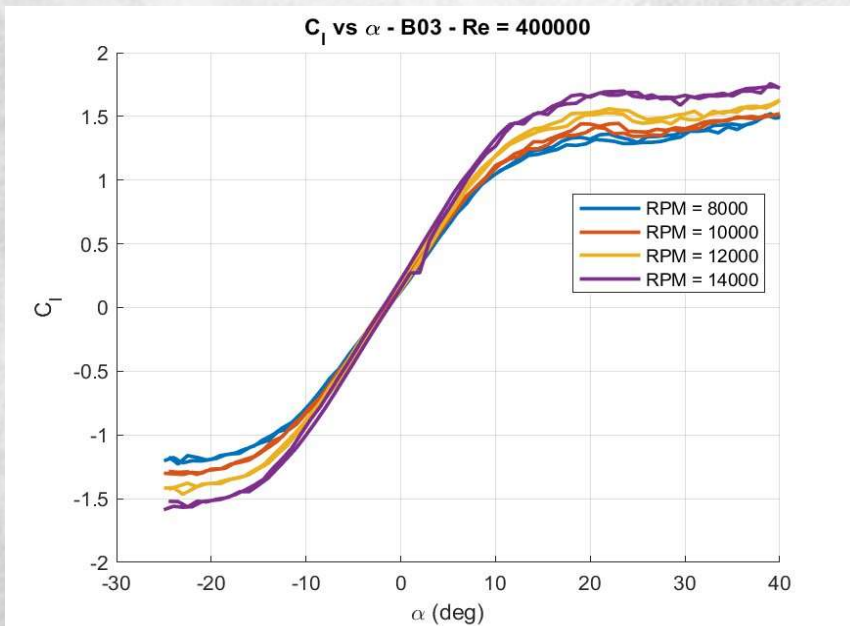
Wing With Propellers



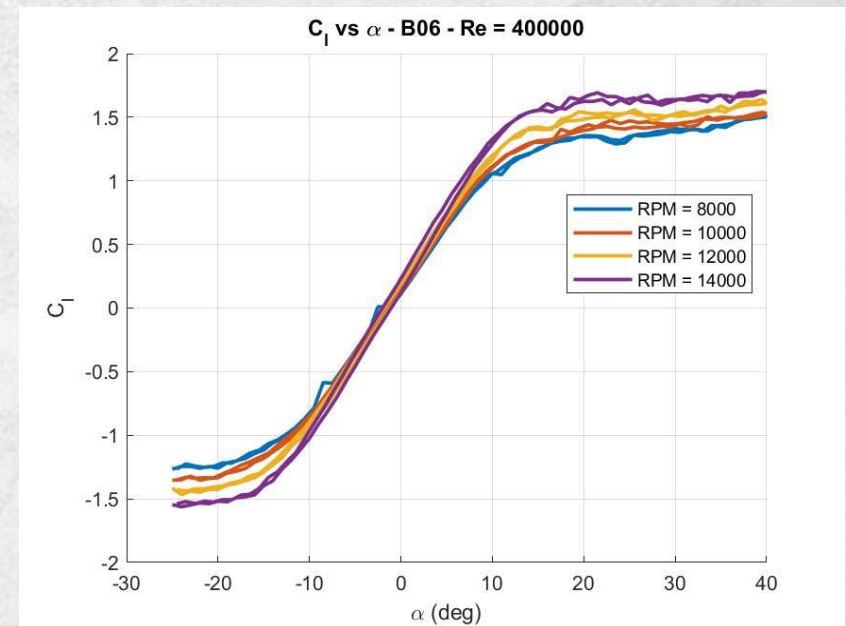
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Results for Cl at 400,000 Reynolds

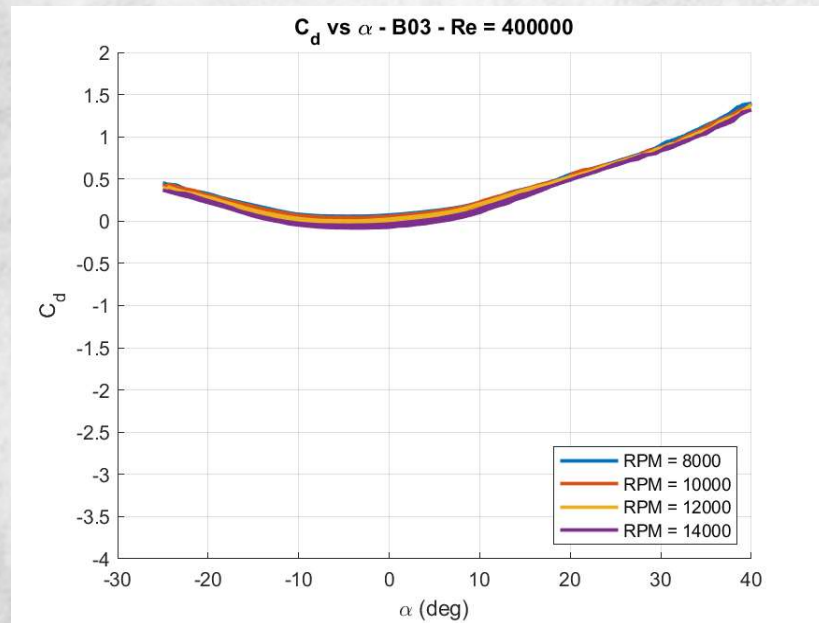


Traditional Leading Edge

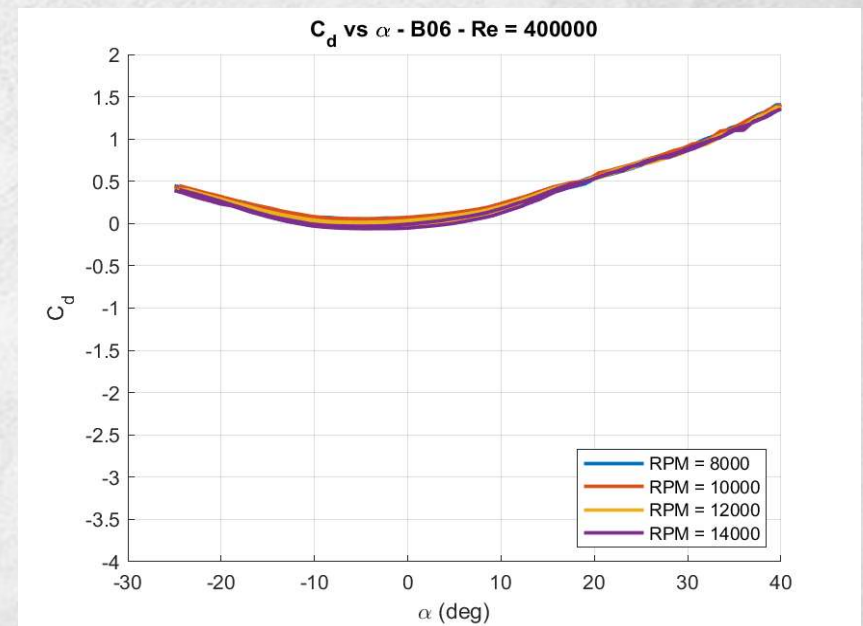


Sinusoidal Leading Edge

Results for C_d at 400,000 Reynolds

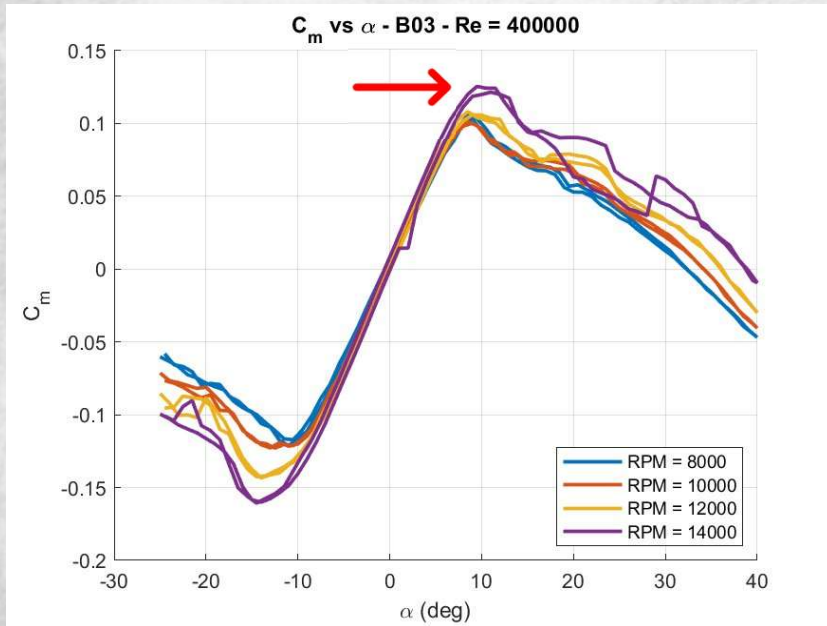


Traditional Leading Edge

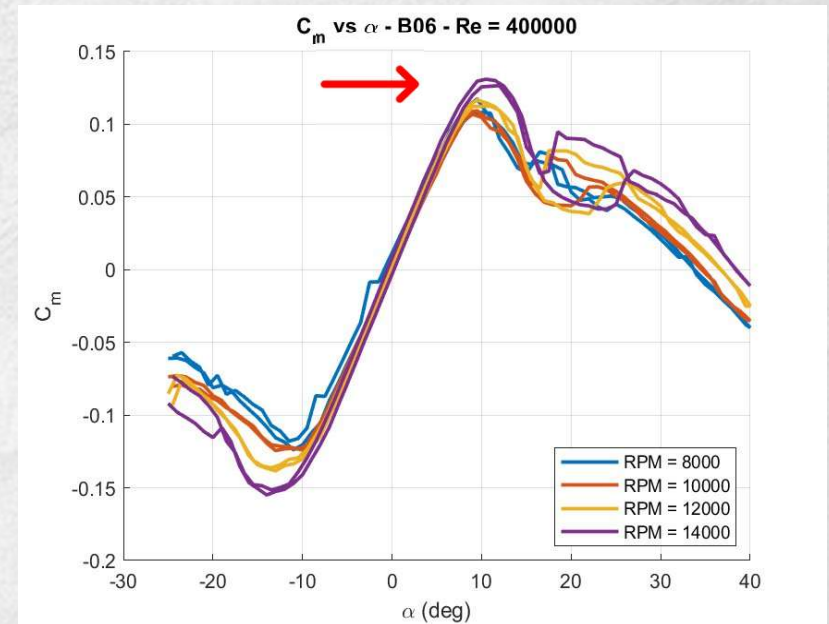


Sinusoidal Leading Edge

Results for C_m at 400,000 Reynolds

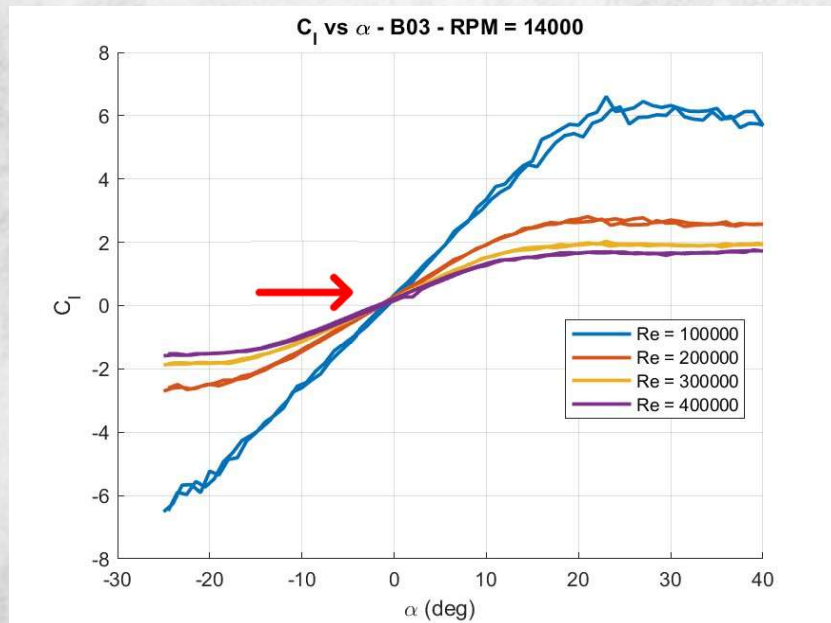


Traditional Leading Edge

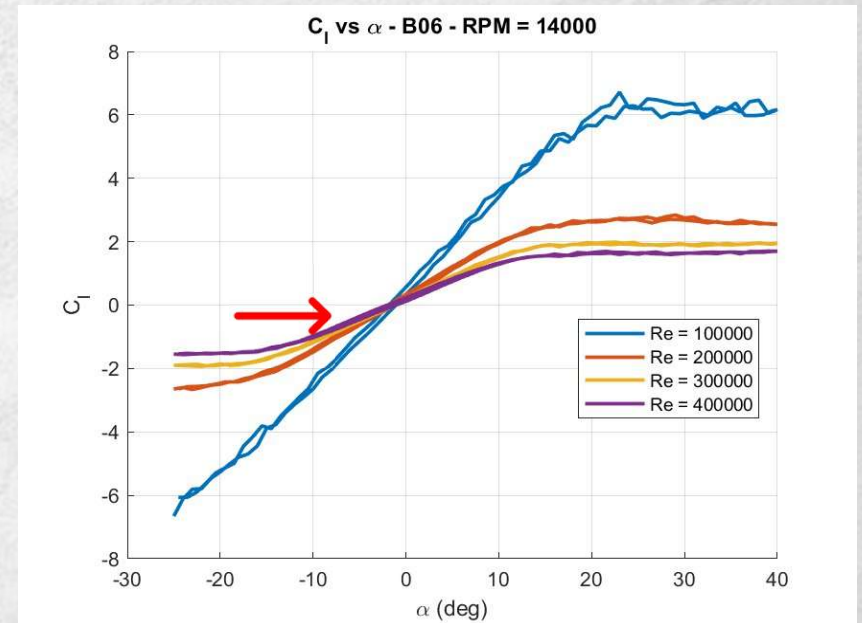


Sinusoidal Leading Edge

Results for CI at 14,000 RPM

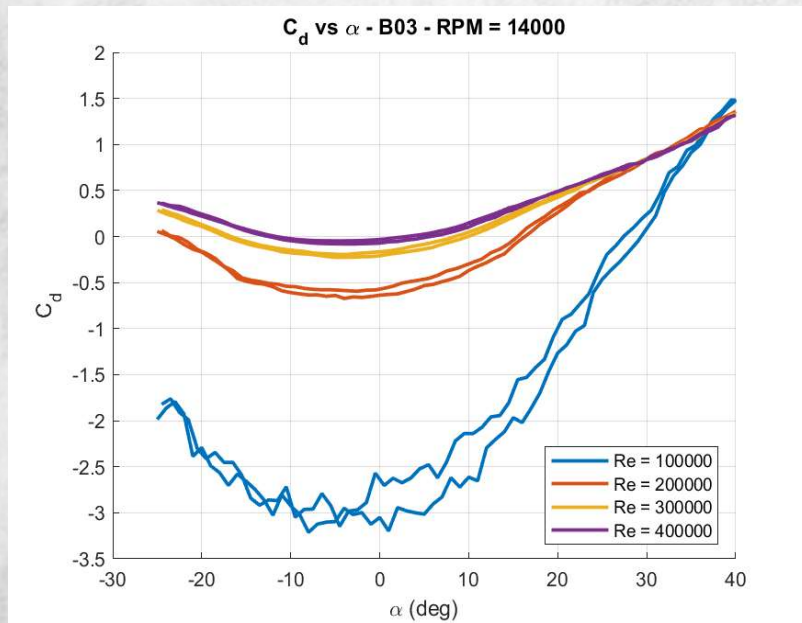


Traditional Leading Edge

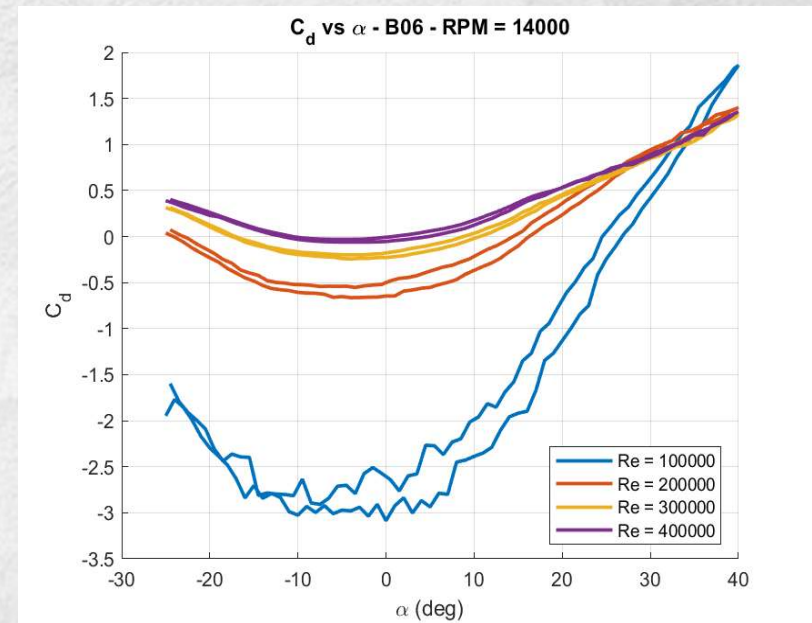


Sinusoidal Leading Edge

Results for C_d at 14,000 RPM

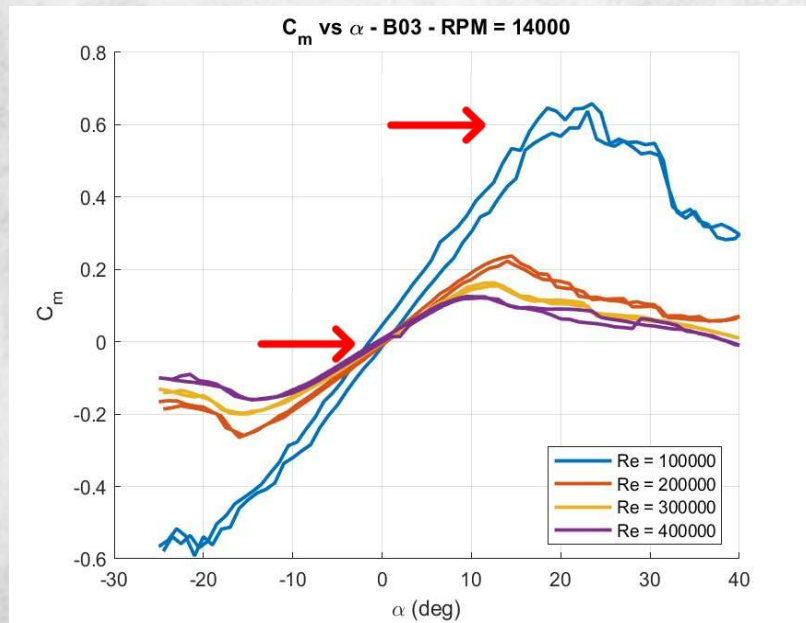


Traditional Leading Edge

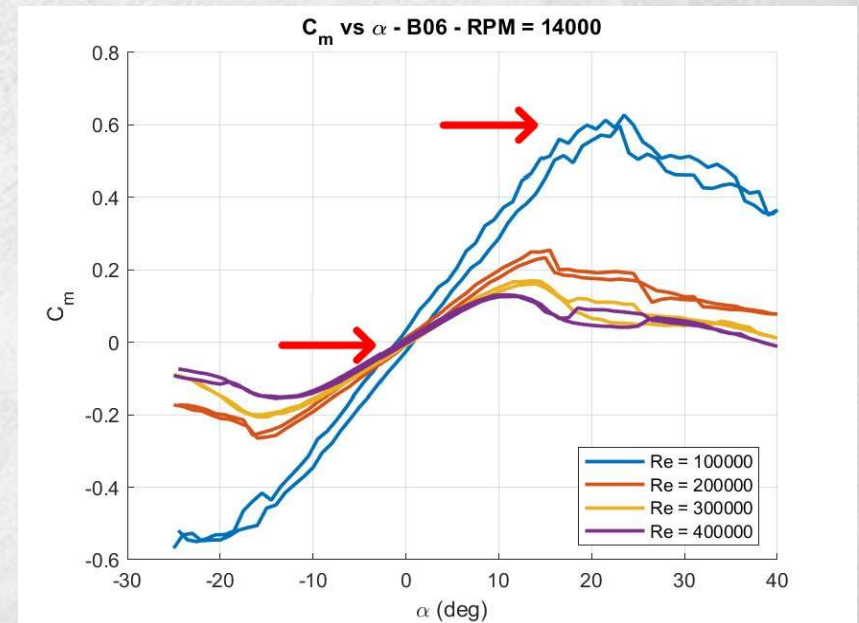


Sinusoidal Leading Edge

Results for C_m at 14,000 RPM



Traditional Leading Edge



Sinusoidal Leading Edge

Observations of Constant Reynolds Number & RPM

- Little to no noticeable changes in coefficients of lift and drag.
 - Without the propellers we had a clear difference, but with them there is almost no difference.
- Setup was stable but had additional vibration throughout wind tunnel tests.

Conclusions

- Our findings support broader applications/research of bio-inspired designs.
- Sinusoidal shapes outperform traditional edges in clean wing studies.
- With propellers attached, data shows no key differences in values due to propulsion dominating the flow behavior.